

**PREDICTIVE QUALITY CONTROL USING**  
**MACHINE LEARNING**  
**IN ADDITIVE MANUFACTURING**

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## **Abstract:**

Additive manufacturing (AM) has become a foundational technology within modern digital factories, enabling the on-demand production of geometrically complex and highly customized components across a wide range of industrial sectors [6, 7, 19]. Despite these advantages, the layer-by-layer fabrication process introduces significant process variability, which can result in defects such as porosity, lack of fusion, warping, delamination, and dimensional inaccuracies [1, 8, 12, 24]. Conventional post-process inspection methods are often limited to defect identification after fabrication and therefore cannot prevent quality degradation during production [1, 24].

To address these limitations, recent research has increasingly focused on integrating machine learning (ML) techniques with in-situ sensing technologies, digital twins, and cyber-physical system (CPS) architectures to enable predictive and closed-loop quality control in AM environments [2, 7, 9, 19, 29]. This report reviews more than thirty recent studies published between 2025 and 2026, covering key developments in sensor fusion, vision- and thermal-based defect detection, melt-pool monitoring, process parameter optimization, digital-twin-enabled process control, predictive maintenance, and explainable artificial intelligence (XAI) for manufacturing applications [2, 4, 13, 14, 17, 18, 20, 26, 29, 32].

Building on the findings of the reviewed literature, a conceptual cyber-physical architecture for predictive quality control in additive manufacturing is proposed. The architecture illustrates the flow of information from physical sensing systems through edge-based data processing and machine learning inference to digital twin platforms that provide feedback for real-time process adjustment and actuator-level control [2, 9, 19, 29]. Finally, the report identifies several critical challenges that continue to hinder the widespread implementation of autonomous quality control systems, including limited availability of high-quality training data, difficulties in model generalization across different machines and materials, real-time computational constraints, and cybersecurity concerns associated with increasingly connected AM infrastructures [5, 9, 13, 24]. The report concludes by outlining future research directions aimed at achieving fully autonomous, self-correcting, and intelligent digital manufacturing environments [1, 2, 9, 29].

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- [1] *AI-based Defect Detection and Self-Healing in Metal Additive Manufacturing* (2025) – <https://doi.org/10.1080/17452759.2025.2515240>
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  - [5] *AI-driven Predictive Maintenance for Industry 4.0: A Systematic Review* (2025/2026) – <https://doi.org/10.1007/s00170-026-17531-w>
  - [6] *Advancing Intelligent Additive Manufacturing* (2025) – <https://doi.org/10.36922/IJAMD025130010>
  - [7] *Current Approaches to Digital Twins in Additive Manufacturing* (2025) – <https://doi.org/10.1007/s40964-025-01262-7>
  - [8] *Detecting the Extent of Co-existing Anomalies in Additively Manufactured Metal Matrix Composites* (2025) – <https://doi.org/10.1080/17452759.2025.2515240>
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  - [18] *In-Situ Porosity Monitoring in Laser Powder Bed Fusion* (2025) – <https://doi.org/10.1016/j.optlastec.2025.111413>
  - [19] *Integration of Machine Learning and Digital Twin in Additive Manufacturing* (2025) – <https://doi.org/10.1007/s40964-025-01257-4>
  - [20] *Integrating AI and IoT for Predictive Maintenance* (Information, 2025) – <https://doi.org/10.3390/info16090737>
  - [24] *Machine Learning Assisted Quality Control in Metal Additive Manufacturing* (Review, 2025) – <https://doi.org/10.1016/j.lmf.2025.00788>
  - [26] *Multimodal Deep Learning for Real-Time Defect Recognition* (2025) – <https://doi.org/10.1016/j.jimse.2025.273>
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  - [32] *Deep-Learning-Based Multisensor Fusion for In-Situ Defect Prediction* (SPIE, 2025) – <https://doi.org/10.1117/12.3063312>

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### **List of Abbreviations:**

|         |                                                         |
|---------|---------------------------------------------------------|
| AM      | Additive Manufacturing                                  |
| CPS     | Cyber-Physical System                                   |
| CPPS    | Cyber-Physical Production System                        |
| CNN     | Convolutional Neural Network                            |
| CT      | Computed Tomography                                     |
| DED     | Directed Energy Deposition                              |
| DT      | Digital Twin                                            |
| FDM/FFF | Fused Deposition Modelling / Fused Filament Fabrication |
| GAN     | Generative Adversarial Network                          |
| IoT     | Internet of Things                                      |
| LPBF    | Laser Powder Bed Fusion                                 |
| LSTM    | Long Short-Term Memory                                  |
| LWIR    | Long-Wave Infrared                                      |
| ML      | Machine Learning                                        |
| NDT     | Non-Destructive Testing                                 |
| NIR     | Near-Infrared                                           |
| PdM     | Predictive Maintenance                                  |
| PINN    | Physics-Informed Neural Network                         |
| WAAM    | Wire-Arc Additive Manufacturing                         |
| XAI     | Explainable Artificial Intelligence                     |
| YOLO    | You Only Look Once (object detection architecture)      |

## **1. Introduction:**

The concept of the digital factory is founded on the integration of physical production assets with computational models, sensor networks, and intelligent control systems, creating what is commonly referred to as a cyber-physical production system (CPPS) [2, 9, 20]. Within this framework, additive manufacturing (AM), commonly known as three-dimensional (3D) printing, has emerged as a key enabling technology because the layer-by-layer fabrication process allows production activities to be continuously monitored, modelled, and adjusted in near real time [6, 7, 19]. This capability aligns closely with the objectives of Industry 4.0, where data-driven decision-making and autonomous process control are central to achieving higher levels of manufacturing efficiency and quality [2, 20].

Unlike conventional subtractive or formative manufacturing processes, AM constructs components incrementally by depositing material layer upon layer. While this approach enables unprecedented design freedom and customization, it also increases sensitivity to process disturbances [6, 24]. Variations such as laser power fluctuations, nozzle clogging, thermal instabilities, irregular powder spreading, and material inconsistencies can affect the quality of individual layers and may propagate throughout the build process [1, 12, 17, 18]. If left undetected, these deviations can accumulate across hundreds or even thousands of layers, ultimately leading to defects that become apparent only after the component has been completed [1, 8, 24]. Consequently, effective in-situ monitoring and predictive quality control have become critical research priorities in the advancement of intelligent additive manufacturing systems [17, 18, 29, 32].

### **1.1. Background:**

Quality assurance in additive manufacturing has traditionally depended on post-process inspection techniques, including computed tomography (CT), dye-penetrant testing, and destructive mechanical testing to assess the structural integrity of manufactured components [24]. Although these methods can effectively identify defects, they are often expensive, time-consuming, and inherently reactive because defects are detected only after the manufacturing process has been completed [1, 24]. As a result, defective builds may lead to significant losses in material, machine utilization, production time, and energy consumption [6, 24].

To overcome these limitations, the digital factory paradigm promotes a shift from reactive inspection to predictive quality control, where manufacturing systems continuously monitor process conditions and anticipate quality deviations before they become critical [2, 7, 9]. In this approach, machine learning models analyze data streams acquired from in-situ sensing technologies, including optical imaging, thermal monitoring, acoustic emission sensors, melt-pool measurements, and other multimodal sensor systems [12, 17, 18, 21, 26, 32]. By identifying patterns associated with defect formation in real time, these models can estimate the likelihood of quality failures during the build process and provide actionable feedback to cyber-physical control systems [1, 29, 32]. Such closed-loop architectures enable dynamic process adjustments, including modifications to process parameters, energy input, or material deposition strategies, thereby reducing defect occurrence and improving overall manufacturing reliability and efficiency [1, 2, 19, 29].

## **1.2. Problem Statement:**

Despite the significant potential of ML-driven quality control, the integration of in-situ sensing, edge computing, digital twins, and actuator feedback into coherent, industrially deployable additive manufacturing systems remains incomplete [2, 7, 9, 19]. Although recent studies have demonstrated substantial progress in process monitoring, defect prediction, and closed-loop control, many solutions remain at the research or pilot stage rather than being fully implemented in industrial production environments [6, 24, 29].

Key barriers include data scarcity, the lack of standardized benchmarking frameworks, and limited model generalizability across different machines, materials, and operating conditions [15, 24, 28]. In addition, real-time computational requirements pose challenges when processing large volumes of multimodal sensor data for immediate decision-making and process control [29, 32]. The increasing connectivity of AM systems through digital twins, cloud platforms, and Industrial Internet of Things (IIoT) technologies also introduces cybersecurity concerns related to data integrity, system reliability, and intellectual property protection [9, 20]. Addressing these challenges is essential for the widespread adoption of autonomous and predictive quality-control systems within future digital factories [2, 7, 19].

## **1.3. Research Objective:**

This report aims to provide a structured review of current machine learning approaches for predictive quality control in additive manufacturing, situate them within the broader framework of cyber-physical systems and digital factories, and synthesize the findings into a conceptual architecture that demonstrates how these increasingly mature technologies can be integrated into a coherent closed-loop manufacturing system [2, 7, 9, 19].

## **1.4. Research Questions:**

The following research questions guide the analysis: (1) Which machine learning techniques and sensing modalities are most effective for in-situ defect prediction in additive manufacturing? (2) How does the digital twin concept operationalize the cyber-physical feedback loop for quality control? (3) What are the principal barriers to the deployment of closed-loop predictive quality-control systems in industrial additive manufacturing environments?

## **1.5. Methodology Overview:**

The report is based on a systematic review of peer-reviewed journal articles, conference proceedings, and preprints published primarily between 2025 and early 2026. Sources were identified through database searches using combinations of the keywords additive manufacturing, machine learning, quality control, digital twins, and cyber-physical systems. More than thirty primary sources were analyzed and synthesized to identify current trends, technological developments, and research challenges in predictive quality control for additive manufacturing.

## 1.6. Structure of the Paper:

The report is organized as follows. Section 2 introduces additive manufacturing processes and the quality challenges that motivate predictive control. Section 3 examines additive manufacturing within the context of cyber-physical systems and digital factory architecture. Section 4 reviews machine learning approaches for predictive quality control, organized according to sensing modality and application area. Section 5 proposes a conceptual integration architecture, while Section 6 discusses current challenges and future research directions. Finally, Section 7 presents the conclusions of the report.

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[32] *Deep-Learning-Based Multisensor Fusion for In-Situ Defect Prediction* (SPIE, 2025) – <https://doi.org/10.1117/12.3063312>

## 2. Additive Manufacturing and the Quality Control Challenge:

### 2.1. Overview of Additive Manufacturing Processes:

Additive manufacturing encompasses a family of processes that build components through the successive deposition of material guided by a digital model [6, 24]. The most industrially significant categories include material extrusion (Fused Deposition Modelling/Fused Filament Fabrication, FDM/FFF), powder bed fusion (Laser Powder Bed Fusion, LPBF, and Electron Beam Melting), and directed energy deposition (DED), including Wire Arc Additive Manufacturing (WAAM) [6, 24]. Each process generates a distinct set of measurable signals: extrusion-based systems produce motion, temperature, and acoustic signatures at the nozzle, whereas powder bed fusion and DED processes generate melt-pool radiation, plume and spatter behavior, and acoustic emissions associated with laser–material interactions [17, 18, 21, 29].

### 2.2. Common Defect Modes

Across these processes, recurring defect categories threaten part quality and certification, including porosity and lack-of-fusion voids caused by insufficient energy density or unstable melt pools; keyhole porosity resulting from excessive energy input; warping, delamination, and residual stresses induced by uneven thermal gradients; dimensional and geometric deviations accumulated layer by layer; surface roughness variations; and spatter- or balling-induced surface defects [1, 6, 12, 18, 24]. In metal powder bed fusion processes, research has shown that porosity formed in one layer can be

inherited, modified, or eliminated by the melt-pool dynamics of subsequent layers, highlighting the highly coupled and time-dependent nature of defect formation [17, 18, 29].

### **2.3. Limitations of Traditional Quality Assurance**

Conventional quality assurance in additive manufacturing relies heavily on ex-situ techniques, including CT scanning, ultrasonic non-destructive testing (NDT), and mechanical testing of witness coupons [24]. While effective for qualification and certification purposes, these methods cannot intervene during the build process, are costly for high-throughput production environments, and provide limited insight into the underlying causes of defects within the complex, multi-physics nature of AM processes [1, 6, 24]. As additive manufacturing continues to evolve from a prototyping technology to a production method for certified aerospace, medical, and automotive components, the demand for in-process, predictive, and closed-loop quality assurance has become a major research priority [6, 7, 24, 29]. This challenge represents a key motivation for the integration of machine learning, advanced sensing technologies, digital twins, and cyber-physical systems within modern manufacturing environments [2, 9, 19, 29].

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- [1] *AI-based Defect Detection and Self-Healing in Metal Additive Manufacturing* (2025) – <https://doi.org/10.1080/17452759.2025.2515240>  
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 [24] *Machine Learning Assisted Quality Control in Metal Additive Manufacturing (Review, 2025)*

## **3. Additive Manufacturing within Cyber-Physical Systems and the Digital Factory**

### **3.1. Cyber-Physical Production Systems**

A cyber-physical production system (CPPS) integrates sensing, computation, communication, and control to establish a continuous feedback loop between the physical manufacturing process and its digital representation [2, 9, 20]. In additive manufacturing, this loop typically comprises: (1) a sensor layer consisting of cameras, pyrometers, photodiodes, microphones, accelerometers, and current/voltage sensors; (2) an edge-computing layer that performs initial signal processing and feature extraction close to the machine to meet real-time latency requirements; (3) a cloud or fog layer that hosts trained machine learning models, historical databases, and digital twin simulations; and (4) an actuation layer that adjusts process parameters, such as laser power, scan speed, extrusion temperature, or deposition path, in response to model predictions [2, 7, 19, 20, 29].

### **3.2. Digital Twins as the Cyber-Physical Bridge**

The digital twin (DT) concept—a continuously updated virtual representation of a physical AM system—is widely regarded as the key mechanism through which cyber-physical integration is achieved [2, 7, 9, 19]. Recent reviews indicate that AM digital twins



have evolved from static simulation models into AI-enhanced, self-optimizing systems capable of supporting real-time monitoring, predictive analytics, and autonomous decision-making across machine, cell, shopfloor, and enterprise levels of the manufacturing hierarchy [2, 7, 9].

In additive to manufacturing, digital twins increasingly combine physics-based thermal and mechanical models with data-driven machine learning components, IoT connectivity, and, in some implementations, advanced visualization environments to provide operators with timely and interpretable feedback on part quality during the build process [9, 11, 19]. This integration enables a more comprehensive representation of process behaviors and supports data-driven quality-control decisions in real time [2, 19].

### **3.3. Industry 4.0/5.0 Context and Cybersecurity**

The digital factory context also introduces new challenges specific to connected AM systems. As AM machines become increasingly networked and instrumented, they may become vulnerable to cyber-physical attacks that manipulate process parameters or sensor data to induce hidden defects or compromise product quality [9, 20]. Recent reviews of digital twins, AI, and cybersecurity in additive manufacturing emphasize that secure data management, side-channel monitoring, and trustworthy AI—including explainable AI (XAI)—should be treated as integral components of cyber-physical system design rather than as afterthoughts [9, 13, 14]. Ensuring the reliability, transparency, and security of these systems is therefore essential for the safe deployment of intelligent and autonomous manufacturing environments [9, 20].

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- [2] *AI-Driven Digital Twins for Manufacturing (Sensors, 2025/2026)* – <https://www.mdpi.com/1424-8220/26/1/124>  
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 [20] *Integrating AI and IoT for Predictive Maintenance (Information, 2025)* – <https://doi.org/10.3390/info16090737>

## **4. Machine Learning Approaches for Predictive Quality Control**

This section reviews recent machine learning approaches for predictive quality control in additive manufacturing, organised around the principal sensing modalities and application areas reported in the 2025–2026 literature. These include in-situ sensor fusion and process monitoring, vision- and thermal-based defect detection, melt-pool monitoring and porosity prediction in metal AM, process parameter optimisation in polymer and extrusion-based AM, digital-twin-integrated machine learning, and predictive maintenance supported by explainable AI [6, 9, 24, 29].

### **4.1. In-Situ Process Monitoring and Multimodal Sensor Fusion**

A major area of current research focuses on the fusion of multiple in-situ sensing modalities—including visible-light, near-infrared (NIR), and long-wave infrared (LWIR) cameras, acoustic sensors, and photodiodes—to predict defects with greater confidence than can be achieved using a single sensor alone [17, 26, 29, 32]. Supervised deep-learning approaches for LPBF have aligned multi-spectral in-situ imaging data with CT-derived porosity labels in the build coordinate system, demonstrating how sensor-fusion pipelines can generate the labelled datasets required for predictive quality models [8, 18, 32].

Complementary research on redundancy analysis in ML-based AM process monitoring has shown that many sensor channels contain overlapping information and has proposed methods to identify and mitigate this redundancy, improving computational efficiency while maintaining predictive performance [28].

Reviews of multimodal sensing and closed-loop control in LPBF further indicate that deep-learning architectures combining melt-pool images, plasma plume signals, and acoustic data are increasingly being used not only for process monitoring but also to support closed-loop adjustment of laser parameters during the build process, representing a practical implementation of the cyber-physical feedback loop discussed in Section 3 [17, 29, 32].

More broadly, recent reviews covering polymer-, metal-, ceramic-, and carbon-based AM systems identify multimodal sensor fusion, in-situ monitoring, and cloud-enabled predictive analytics as key enablers of autonomous and intelligent additive manufacturing environments [6, 24].

## **4.2. Vision-Based and Thermal Deep Learning Defect Detection**

Vision-based defect detection using convolutional neural networks (CNNs) and modern object-detection architectures remains one of the most mature applications of machine learning in additive manufacturing quality control [16, 24, 31]. Recent studies have demonstrated the use of deep-learning-based frameworks trained on large, labelled image datasets to enable real-time, in-situ defect identification during the fabrication process [16, 24].

Thermal sense combined with deep learning has also emerged as a prominent research area. Studies on laser-based powder bed fusion have explored CNNs, long short-term memory (LSTM) networks, generative adversarial networks (GANs), and physics-informed neural networks (PINNs) for process monitoring, highlighting the complementary strengths of these approaches. CNNs provide strong spatial feature extraction, LSTMs capture the temporal evolution of defects, GANs help address data scarcity through synthetic data generation, and PINNs improve model generalization by incorporating physical process knowledge [12, 29, 31]. However, challenges related to labelled-data availability and real-time deployment remain significant [24, 29].

A conditional autoencoder with skip connections combined with in-situ infrared sensing has framed defect detection as a reconstruction-based anomaly-detection problem, producing a continuous anomaly severity score rather than a binary classification outcome. This approach is particularly well suited to predictive quality-control applications because it enables the early identification of process deviations before critical defects develop [12].

For multi-material and composite AM applications, explainable multi-camera fusion approaches combining dual mid-wave infrared imaging with explainable feature-selection techniques have been proposed to detect and interpret co-existing anomalies of varying severity in metal matrix composites [8]. More broadly, reviews of AI-based defect detection and self-healing in metal additive manufacturing highlight how defect-detection outputs can be coupled with in-situ repair or re-melting strategies, extending quality-control systems from defect identification towards autonomous process correction [1, 24].

### **4.3. Melt-Pool Monitoring and Porosity Prediction in Metal Additive**

#### **Manufacturing**

Porosity remains one of the most extensively studied defects in metal powder bed fusion due to its direct influence on mechanical performance, fatigue life, and structural reliability [18, 24]. A spatiotemporal porosity prediction model combining a CNN for intra-layer spatial feature extraction with a bidirectional LSTM for modelling cross-layer temporal dependencies has been proposed, motivated by experimental evidence showing that lack-of-fusion pores may be inherited, modified, or eliminated across successive layers depending on melt-pool dynamics [17, 18]. This approach highlights the importance of capturing both spatial and temporal process behavior when predicting defect formation in LPBF systems.

A related approach, termed AM Net, employs a self- and cross-attention neural network to fuse static and dynamic melt-pool features extracted from high-speed imaging data, reporting prediction accuracies exceeding 99% while maintaining inference times suitable for real-time deployment. The model also demonstrated strong performance when trained on a reduced proportion of the available dataset, indicating its potential for data-efficient quality monitoring applications [18].

Reviews of multimodal sensing and closed-loop control in LPBF compare infrared thermography, acoustic emission sensing, and optical melt-pool imaging for porosity prediction and process monitoring [17, 29]. These studies further discuss how predictive signals can be integrated into closed-loop control strategies that dynamically adjust laser power and scan speed during fabrication [29, 32]. Collectively, this body of research demonstrates a growing shift towards physics-informed and attention-based learning architectures that seek to balance predictive accuracy with the computational efficiency required for real-time, layer-by-layer quality control [18, 29].

### **4.4. Process Parameter Optimization and Quality Prediction in Extrusion-Based AM**

For material extrusion (FDM/FFF), a substantial body of recent research applies machine learning to map process parameters—including layer thickness, infill density, nozzle and bed temperature, print speed, and shell count—to part quality metrics such as dimensional accuracy, surface roughness, and mechanical performance [22, 25, 27]. A recent review of FDM process parameter optimization highlights the growing use of deep learning for predicting forming accuracy and reinforcement learning for adaptive, real-time process control, while noting that multi-objective optimization for complex geometries and composite materials remains an important area for further research [30].

Several recent studies provide concrete examples. A multi-objective optimization study combining a full-factorial design of experiments with a Random Forest regressor reported substantial improvements in the prediction of tensile strength, elastic modulus, and strain at maximum stress compared with conventional response-surface methodologies [25].

Similarly, research applying ensemble learning methods—including Random Forest, J48 decision trees, Naïve Bayes, and multilayer perceptron's—to predict the surface quality of polylactic acid (PLA)-based components identified infill density and layer thickness as the dominant factors affecting hardness, tensile strength, flexural strength, and surface roughness [27]. Work on attention-mechanism-enhanced multimodal convolutional neural networks (ATT-MM-CNN) further demonstrated that integrating multiple input modalities with attention mechanisms can significantly improve tensile-

property prediction accuracy compared with conventional machine-learning approaches [10].

Additional studies have explored porosity prediction in FDM using pre-trained and transferable machine-learning models to reduce the experimental burden associated with process development [23]. Other research has investigated the combined effects of layer thickness, infill density, and shell count on printing time, surface quality, and dimensional accuracy, highlighting the trade-offs that must be managed when optimizing extrusion-based additive manufacturing processes [27].

#### **4.5. Digital-Twin-Integrated Machine Learning**

A systematic literature review of digital twins in additive manufacturing identifies the need for more accurate material-behavior and structural-analysis models to support efficient simulation environments and notes that the integration of these models with data-driven machine learning components remains limited in practice [7].

A complementary review focused on polymer-based additive manufacturing describes how digital twins combined with machine learning, for example, through the use of grey relation analysis and Random Forest models to optimize printing parameters for bio-based materials—can support both process optimization and real-time quality prediction. The review also discusses cyber-physical production system implementations based on IoT technologies and the 5C (Connection–Conversion–Cyber–Cognition–Configuration) architecture for large-format 3D printing [19].

At a broader manufacturing-systems level, a review of AI-driven digital twins proposes a dual-axis taxonomy that maps AI techniques across different levels of the manufacturing hierarchy (machine, cell, shopfloor, and enterprise) and across functional categories, including monitoring, prediction, optimization, control, and implementation. This framework provides a useful basis for positioning AM-specific predictive quality-control systems within the wider digital factory context [2].

Finally, a comprehensive review of digital twins, AI, and cybersecurity in additive manufacturing concludes that, although digital twins, AI technologies, and cybersecurity solutions are individually advancing, their integrated application remains fragmented. The authors therefore propose a future research framework that explicitly links predictive machine-learning models with secure, real-time digital twin feedback loops [9].

#### **4.6. Predictive Maintenance, Acoustic Monitoring, and Explainable AI**

Predictive quality control in the digital factory is closely linked to predictive maintenance (Pd.M.) of manufacturing equipment, as machine-level degradation, including nozzle wear, contamination of laser optics, and motor or bearing faults—can directly contribute to part-level quality defects [3, 4, 5, 20]. A systematic review of AI-driven predictive maintenance in Industry 4.0 classifies Pd.M. approaches into physics-based, knowledge-based, data-driven, and hybrid categories, and identifies real-time deployment, model robustness, heterogeneous data management, and cybersecurity as key remaining challenges [5].

A practical study integrating AI and IoT technologies for predictive maintenance demonstrates temperature- and vibration-based prediction models for electric motors that are directly applicable to additive manufacturing machine subsystems [20]. Acoustic monitoring also continues to attract attention as a cost-effective complement to optical and thermal sensing technologies. In wire arc additive manufacturing (WAAM), a recent study proposed contact acoustic-emission-based in-situ anomaly detection using deep

unsupervised learning, showing that the combination of frequency-domain energy features, convolutional autoencoders, and Isolation Forest algorithms substantially improve anomaly-detection performance compared with conventional control-chart and manually engineered feature approaches [21].

A related multimodal framework integrates visual, acoustic, and vibration signals for real-time defect recognition, illustrating the growing convergence between additive manufacturing monitoring systems and broader smart-manufacturing condition-monitoring technologies [26]. Explainability has also emerged as a critical requirement for industrial adoption. Research on explainable AI-driven quality and condition monitoring argues that the opaque nature of high-performing deep-learning models remains a significant barrier to industrial trust, validation, and regulatory acceptance, while demonstrating how explainable AI (XAI) techniques can improve model transparency without compromising predictive performance [14].

Similarly, explainable AI-driven predictive maintenance frameworks for safety-critical equipment emphasize that interpretability is essential not only for operator confidence but also for understanding why a model identifies a potential fault or defect. This capability is particularly important for additive manufacturing cyber-physical systems, where effective closed-loop control depends on linking predictive insights to appropriate corrective actions [13].

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## 5. Proposed Conceptual Architecture for Predictive Quality Control

Synthesizing the literature reviewed in Section 4, this section proposes a five-layer conceptual architecture for predictive quality control in additive manufacturing. The architecture is designed to align with the cyber-physical systems framework outlined in Section 3 and to serve as a reference model for future implementation, validation, or simulation studies in intelligent manufacturing environments [2, 7, 9, 19, 29].

## **5.1. Layered Architecture**

The proposed architecture comprises the following five layers:

**Sensing Layer:** Multimodal in-situ sensors—including high-speed visible, NIR, and LWIR cameras, photodiodes, acoustic-emission sensors, microphones, accelerometers, and current and temperature sensors—capture raw process signals synchronized with the build coordinate system [17, 21, 29, 32].

**Edge Processing Layer:** Local computing resources perform signal pre-processing, feature extraction, redundancy reduction, and lightweight inference (e.g., autoencoder-based anomaly scoring) to satisfy the millisecond-to-second latency requirements of layer-wise feedback and process monitoring [12, 28, 29].

**Machine Learning / Prediction Layer:** Trained models—including CNNs and transformer-based architectures for spatial defect classification, LSTM/Bi-LSTM and attention-based networks for temporal and cross-layer dependency modelling, and physics-informed or hybrid models for porosity and thermal-field prediction—generate defect-probability and severity scores at the layer or region level [12, 18, 29, 31].

**Digital Twin / Decision Layer:** A continuously updated digital twin integrates sensor data, machine-learning predictions, and physics-based simulations to assess the predicted impact on final part quality and determine corrective actions, such as adjustments to laser power, scan speed, extrusion temperature, or deposition paths [2, 7, 9, 19].

**Actuation and Cloud/Enterprise Layer:** Corrective commands are transmitted to the AM machine controller to enable real-time process adjustment, while aggregated operational data support enterprise-level analytics, predictive maintenance systems, and explainable-AI dashboards for operator review and continuous model improvement [2, 14, 20].

## **5.2. Closing the Loop:**

The defining characteristic of this architecture, relative to conventional inspection-based approaches, is the bidirectional flow of information between the prediction layer and the actuation layer through the digital twin [2, 7, 9, 29]. Rather than simply identifying a defect after it has occurred, the digital twin evaluates whether a predicted anomaly, for example, an emerging lack-of-fusion region detected from melt-pool features—has the potential to develop into a critical defect. If the predicted impact exceeds an acceptable threshold, the system computes a corrective parameter adjustment that can be applied within the same layer or in a subsequent layer of the build process [18, 29].

This approach relies on redundancy-aware and computationally efficient sensor-fusion strategies to remain viable under real-time operating constraints. By reducing unnecessary data processing while preserving predictive performance, such methods support the rapid decision-making required for closed-loop quality control in additive manufacturing environments [28, 29, 32].

### **5.3.Explainability and Security as Cross-Cutting Concerns:**

Consistent with the literature reviewed in Sections 4.6 and 3.3, explainable AI (XAI) and cybersecurity should be treated as cross-cutting requirements that span all five layers of the proposed architecture [9, 13, 14]. Explainability enables operators and quality engineers to understand, validate, and audit the reasoning behind model predictions and corrective actions, thereby increasing trust, transparency, and accountability in automated quality-control systems [13, 14]. At the same time, secure data management, trustworthy communication infrastructures, and side-channel monitoring mechanisms help protect the integrity of the sensor-to-actuator feedback loop against cyber-physical manipulation, data tampering, and other security threats that could compromise part quality or system reliability [9, 20].

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## **6. Challenges and Future Directions:**

### **6.1.Data Scarcity and Standardization:**

Despite rapid progress, the availability of large, standardized, and well-labelled datasets remains one of the most significant barriers to the wider adoption of machine learning for predictive quality control in additive manufacturing [15, 24, 31]. Recent research has sought to address this challenge by proposing standardized, vision-based data-collection frameworks that align G-code-derived digital references with imaging systems to capture layer-wise dimensional deviations in reproducible, physically meaningful units [15].

Synthetic data generation using generative models and transfer learning from pre-trained networks have emerged as promising strategies for reducing data requirements and improving model development efficiency [22, 31]. However, the ability of these approaches to generalize reliably across different machines, process settings, materials, and production environments remains largely unvalidated at industrial scale, highlighting a critical gap between laboratory demonstrations and real-world deployment [24, 31].

## **6.2. Real-Time Computational Constraints:**

Many high-performing machine-learning architectures—including transformers and attention-based fusion networks—can achieve excellent predictive accuracy, but their computational complexity presents a challenge for real-time deployment in additive manufacturing environments [18, 29, 32]. To satisfy the millisecond-scale latency requirements associated with layer-wise monitoring and melt-pool-level feedback, these models often require careful optimization through lightweight architectures, parameter reduction, efficient feature-selection strategies, or deployment on edge-computing platforms located close to the manufacturing process [28, 29]. Balancing predictive performance with computational efficiency therefore remains a critical requirement for practical closed-loop quality-control systems [24, 29].

## **6.3. Model Generalization and Multi-Material Systems:**

Most published studies are validated using a single machine, material system, or part geometry, limiting confidence in their broader industrial applicability [24, 31]. As additive manufacturing continues to evolve towards multi-material, multi-process, and increasingly interconnected production environments, predictive models will need to generalize across a much wider range of process conditions and heterogeneous sensor signatures [6, 24]. Achieving this level of robustness remains a major research challenge. Digital twin frameworks that combine physics-based process models with data-driven machine-learning components are increasingly viewed as a promising solution, as they may offer greater transferability and adaptability than purely data-driven approaches when deployed across different machines, materials, and operating conditions [7, 9, 19].

## **6.4. Explainability, Trust, and Regulatory Compliance:**

For additive manufacturing applications in highly regulated sectors such as aerospace and medical devices, the adoption of black-box predictive models remains challenging due to stringent requirements for traceability, validation, and certification [14, 24]. The growing emphasis on explainable AI (XAI) for quality and condition monitoring suggests that future predictive quality-control systems will need to provide interpretable evidence linking sensor observations, model predictions, and corrective actions [13, 14]. Such transparency is essential for building operator trust, supporting regulatory compliance, and providing the documented justification required for part qualification and certification in safety-critical manufacturing environments [13, 14].

## **6.5. Cybersecurity of Connected AM Systems:**

As predictive quality control increasingly relies on continuous, networked sensor-to-actuator feedback loops, additive manufacturing cyber-physical systems become potential targets for adversarial manipulation of sensor data, communication channels, or process parameters [9, 20]. Such attacks could compromise by monitoring accuracy, disrupt control decisions, or introducing hidden defects that remain undetected until later stages of production [9]. Consequently, future research should treat cybersecurity as a core design requirement rather than an auxiliary consideration. In particular, the integration of side-channel monitoring, cyber-physical anomaly detection, secure data management, and



trustworthy digital twin synchronization will be essential for maintaining the integrity, reliability, and safety of predictive quality-control architectures [9, 20].

## **6.6. Towards Autonomous Self-Correcting Additive Manufacturing:**

The convergence of multimodal sensor fusion, attention-based and physics-informed deep-learning methods, digital-twin-driven closed-loop control, predictive maintenance, and explainable AI are steadily moving additive manufacturing towards systems capable of autonomously detecting, predicting, and correcting quality issues with minimal human intervention [2, 9, 24, 29]. Collectively, these technologies provide the foundation for more intelligent, adaptive, and self-optimizing manufacturing environments that align closely with the broader vision of the digital factory [2, 6, 9].

Realizing this vision, however, will require continued progress in several key areas, including standardized data infrastructures, computationally efficient and explainable machine-learning models, robust cross-machine and cross-material generalization, and secure cyber-physical integration [9, 14, 15, 24]. Addressing these challenges will be essential to achieving reliable, scalable, and industrially deployable predictive quality-control systems for next-generation additive manufacturing environments [7, 9, 19].

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## **7. Conclusion**

This report has explored how machine learning is reshaping quality control in additive manufacturing within the wider context of digital factories and cyber-physical systems. The literature reviewed from 2025 and early 2026 shows a clear movement away from traditional inspection methods that identify defects only after a build is complete. Instead, research is increasingly focused on predictive and real-time approaches that can detect potential quality issues as they develop and, in some cases, take corrective action before they affect the final part.

Several technologies have emerged as key enablers of this transition. Multimodal sensor fusion, combining optical, thermal, acoustic, and process-monitoring data, has significantly improved the ability to capture the complex behavior of additive manufacturing processes. At the same time, advances in machine learning, particularly through CNNs, LSTMs, attention-based models, transformers, and physics-informed approaches, have strengthened the prediction of defects, porosity, and process instability. Digital twins further enhance these capabilities by creating a continuous link between the physical process and its virtual representation, enabling more informed and responsive decision-making. Predictive maintenance and explainable AI also play an increasingly important role by extending quality assurance beyond the manufactured component itself to include machine condition, system reliability, and user confidence in automated decisions.

One of the strongest observations from this review is that the future of predictive quality control is unlikely to depend on a single breakthrough technology. Rather, the most

promising developments come from the integration of sensing, machine learning, digital twins, and process control into a unified cyber-physical system. While each of these technologies has shown significant progress individually, their true value emerges when they operate together as part of a connected and intelligent manufacturing environment. In many ways, this integration represents the real shift taking place within modern additive manufacturing.

The five-layer architecture proposed in this report reflects this understanding by bringing together sensing, edge processing, machine-learning prediction, digital twin decision-making, and actuation within a single framework. Although the individual building blocks are becoming increasingly mature, their deployment in fully integrated industrial systems remains limited. From my perspective, bridging this gap between research success and industrial implementation is one of the most important challenges that must be addressed in the coming years.

The review also highlights several obstacles that continue to slow adoption. Data scarcity remains a persistent issue, particularly the lack of large, standardized datasets that can support robust model development and comparison. Real-time computational requirements, limited model transferability across machines and materials, and cybersecurity concerns within connected manufacturing environments present additional challenges. While many studies report impressive prediction accuracies under controlled experimental conditions, relatively few demonstrate consistent performance across different production settings. This suggests that future research should focus not only on improving model accuracy but also on enhancing robustness, scalability, and industrial applicability.

The findings of this report provide clear answers to the research questions posed at the outset. First, multimodal sensing combined with attention-based and deep-learning architecture currently appears to offer the most effective approach for in-situ defect prediction. Second, digital twins serve as the mechanism through which the cyber-physical feedback loop is realized by continuously integrating sensor data, machine-learning predictions, and physics-based process knowledge to support corrective actions. Third, the main barriers to industrial deployment remain data availability and standardization, real-time processing requirements, limited generalization across manufacturing environments, explainability, and cybersecurity.

Overall, the evidence reviewed suggests that additive manufacturing is steadily progressing towards intelligent systems capable of monitoring, predicting, and eventually correcting quality issues with minimal human intervention. Although fully autonomous quality control is still some distance away, the direction of current research is both clear and encouraging. In my view, the next major advances will come not from developing increasingly complex machine-learning models alone, but from creating reliable, secure, and interoperable systems that allow these technologies to work together effectively in real manufacturing environments. Achieving this integration will be essential for turning the vision of self-correcting digital factories from a research objective into an industrial reality.

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## 9. **Tools Used**

The preparation of this dissertation involved the use of several digital and AI-assisted tools to support data analysis, numerical comparison, formatting, grammar checking, and language refinement. These tools were strictly supportive, and all technical interpretation, validation, and conclusions were performed by the author, ensuring academic integrity.

### 9.1. **AI-Based Tools**

AI Chatbots (e.g., ChatGPT):

- Used to cross-check numerical consistency across tables, such as lead times, material waste percentages, and cost estimations.
- Assisted in summarizing literature trends, structuring sections, and improving technical clarity.
- All quantitative conclusions were verified manually against original sources to ensure accuracy.

### 9.2. **Numerical Comparison Tools**

Spreadsheet Software (e.g., Microsoft Excel):

- Organized quantitative data from multiple sources.
- Calculated percentage reductions, cost-per-part estimations, material utilization, and energy consumption.
- Generated tables and figures presented in Sections 4.2 (Temporal Efficiency), 4.3 (Material Waste), and 4.5 (Economic Considerations).

### **9.3. Language and Formatting Tools**

#### **1. Grammar and Spelling Checkers (e.g., Grammarly):**

- Improved grammar, sentence structure, and academic tone without altering technical meaning.

#### **2. Paraphrasing and Rewriting Tools:**

- Used selectively to enhance clarity and reduce redundancy while preserving original technical content.

#### **3. Formatting Tools:**

- Applied for consistent headings, table layouts, figure captions, and reference styling according to academic guidelines.

*All tools were applied under **strict academic integrity**, with the author retaining full responsibility for all interpretations, analyses, and conclusions.*